

Brain Stem

Department Anatomy
Neuroanatomy (3)

Second Part

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Lectuer Points:

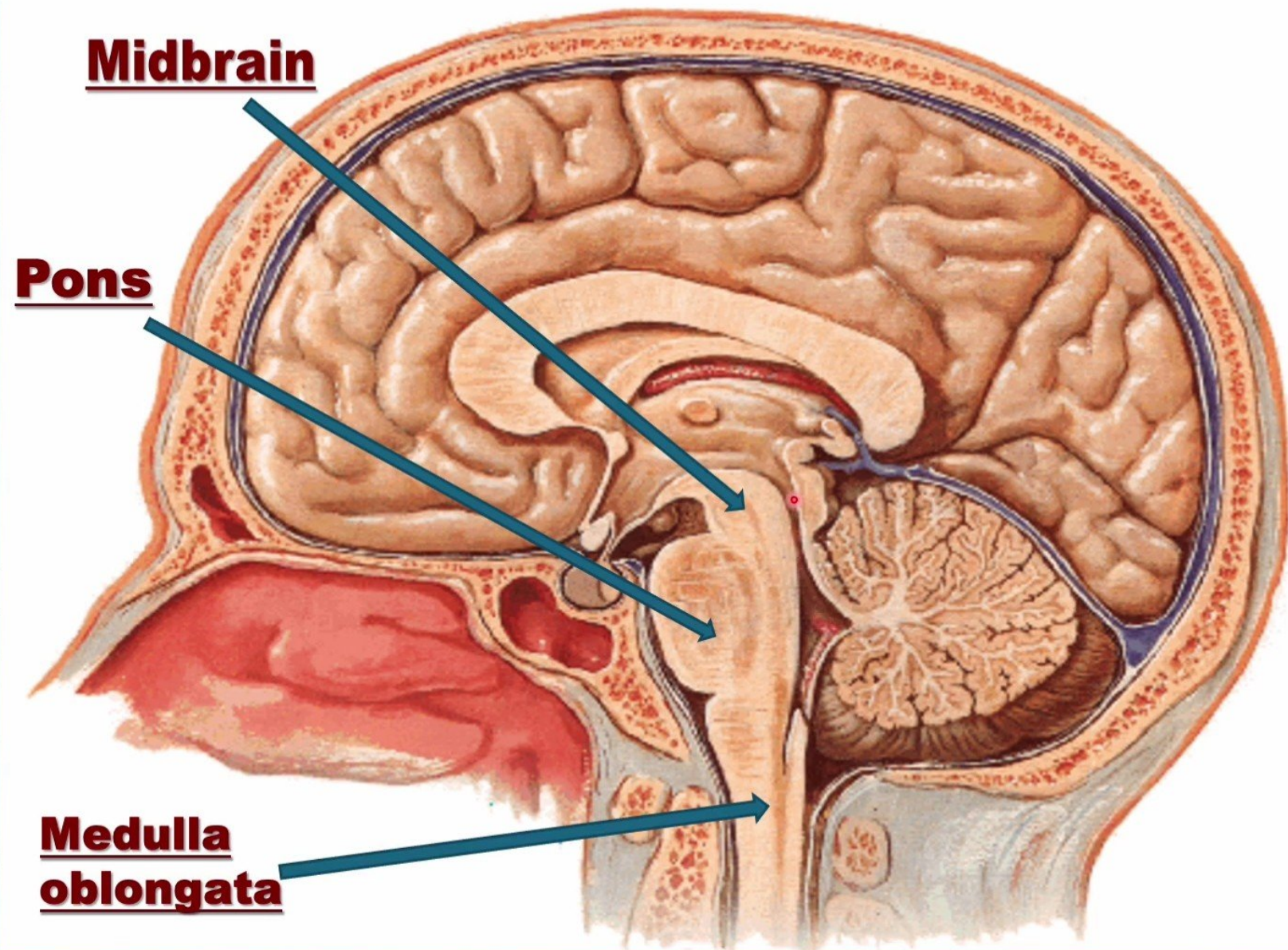
- Part of the brain stem.
- The brainstem has three broad functions.
- The brainstem anterior and posterior view.
- Cranial nerves and Brain stem.
- The medulla oblongata:
 - External features:
 - ✓ The anterior view of medulla oblongata.
 - ✓ The posterior surface of the medulla oblongata.
 - Internal features:
 - ✓ Transverse sections of the medulla oblongata at Level of decussation of the pyramids.
 - ✓ Transverse sections at Level of decussation of the medial lemnisci
 - ✓ Transverse sections at the level of the middle of the olivary nuclei.

- The pons:
 - External features:
 - ✓ Anterior surface of the pons.
 - ✓ Posterior surface of the pons.
 - Internal features:
 - ✓ Transverse section through the caudal part of the pons at the level of the facial colliculus.
 - ✓ Transverse section through the pons at the level of the trigeminal nuclei
- The midbrain :
 - External features:
 - ✓ Anterior surface of the midbrain.
 - ✓ Posterior surface of the midbrain.
 - ✓ Lateral view of the midbrain.
 - Internal features:
 - ✓ Transverse section at the level of the superior colliculus.
 - ✓ Transverse sections of the midbrain at the level of the inferior colliculus.
- Summary of Anterior view of the Brain Stem.
- Summary of Posterior view of the Brain Stem.
- Summary of Cranial nerve nuclei in brain stem.
- Principal arteries of the brain stem (ventral view).
- Corticospinal (pyramidal) tract.

The brain stem

It is made up of:

- The midbrain.
 - The pons.
 - The medulla oblongata.
- It occupies the **posterior cranial fossa** of the skull.



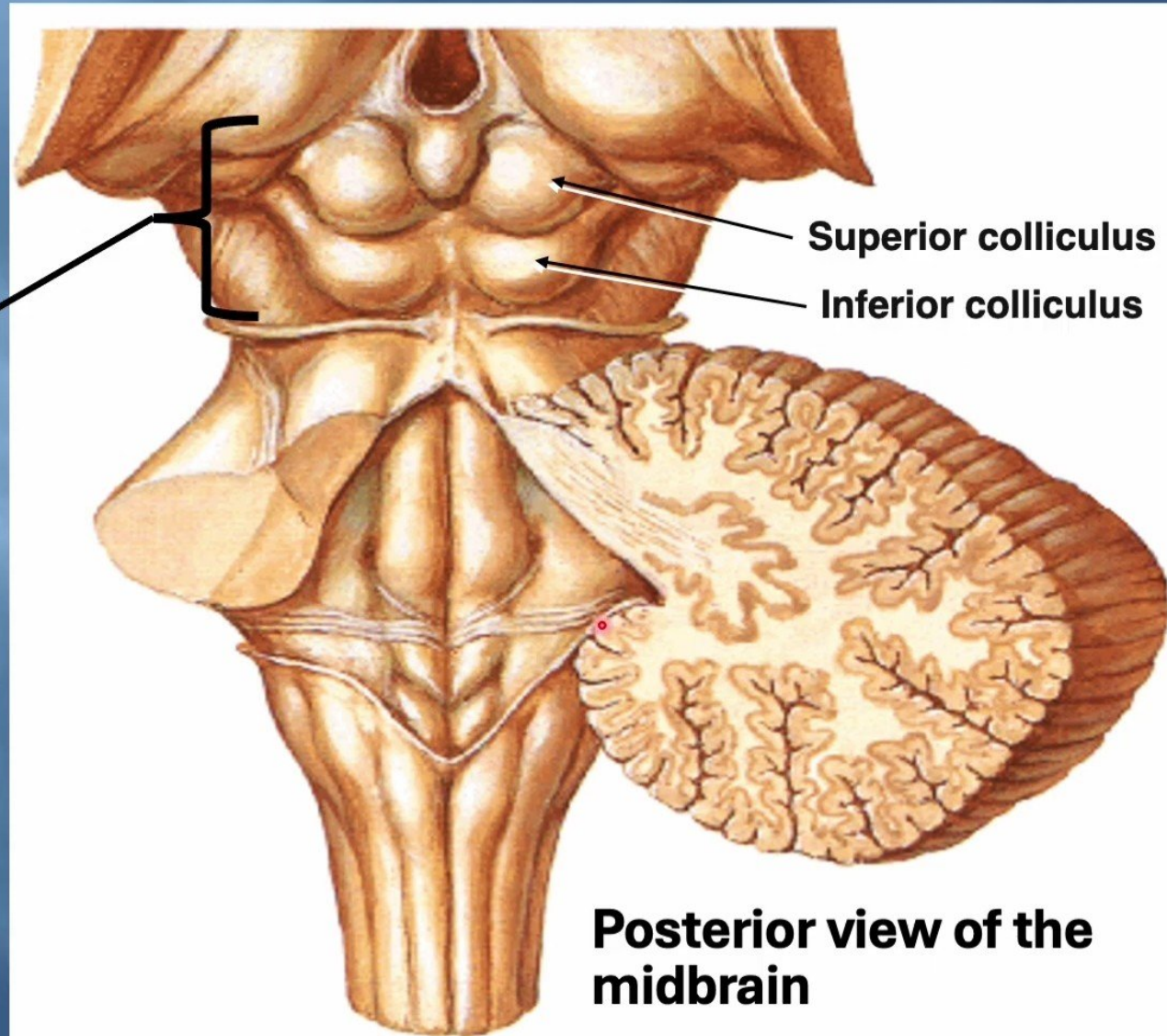
The midbrain

- It is **about 0.8 inch (2 cm)** in length.
- It connects the **pons** and **cerebellum** with the forebrain.
- It ascends through the **opening in the tentorium cerebelli**.

The cerebral aqueduct: is **narrow** channel filled with **cerebrospinal fluid**.

The corpora quadrigemina:

- They are found on the **posterior surface** of midbrain.
- They are **four colliculi** (rounded eminences) divided into **superior** and **inferior** pairs by a **vertical** and a **transverse groove**.
- The **superior colliculi** are centers for **visual reflexes**.
- The **inferior colliculi** are **lower auditory centers**.



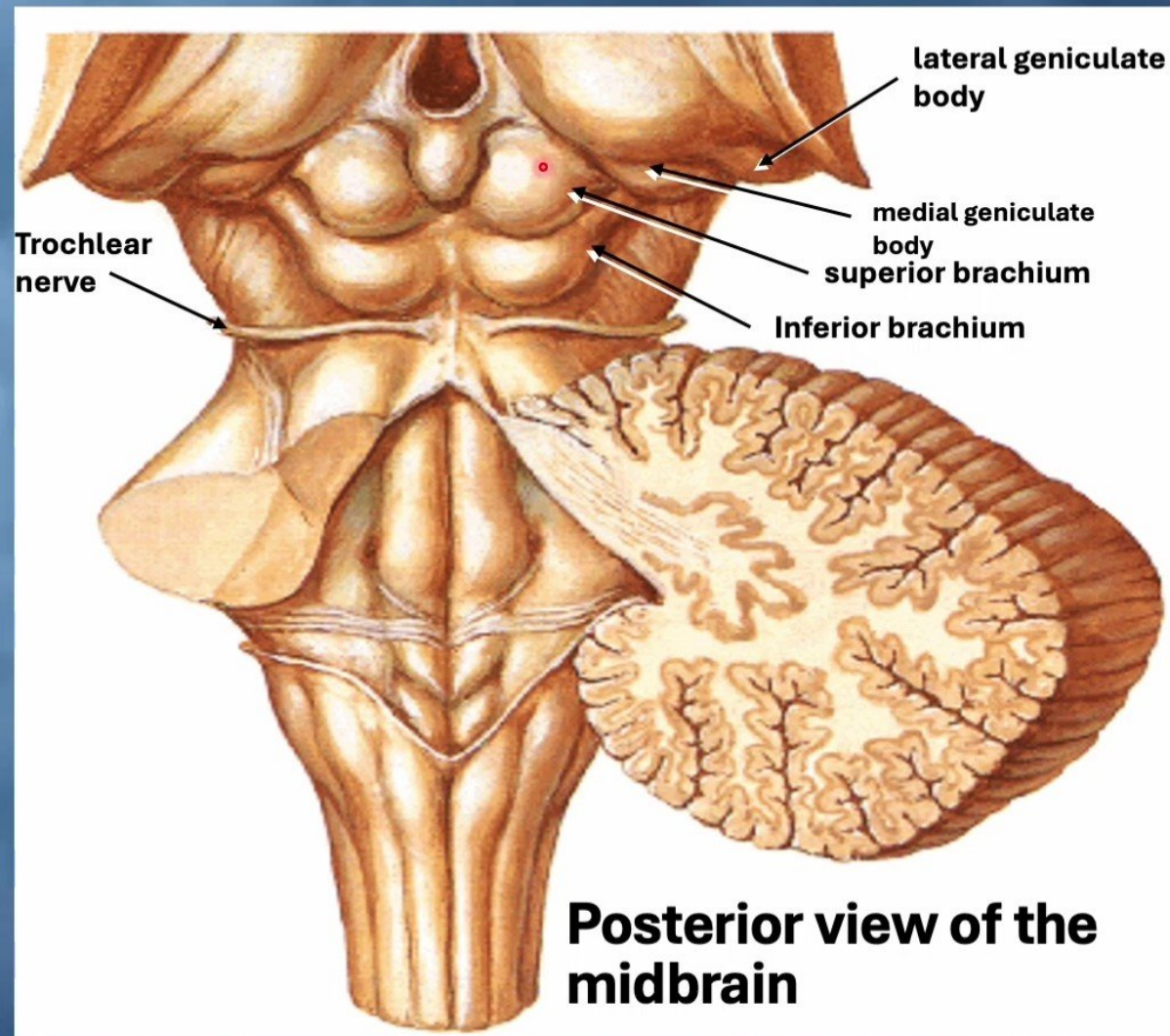
Trochlear nerve:

- ✓ It emerges in the **midline below** the **inferior colliculi**.
- ✓ That **wind around the lateral aspect** of the **midbrain** to enter the lateral wall of the **cavernous sinus**.

The superior and inferior brachium:

- ✓ On the lateral aspect of the midbrain, the **superior** and **inferior brachia** ascend in an anterolateral direction.
- ✓ The **superior brachium** passes from the **superior colliculus** to the **lateral geniculate body** and the **optic tract**.
- ✓ The **inferior brachium** connects the **inferior colliculus** to the **medial geniculate body**.

The midbrain



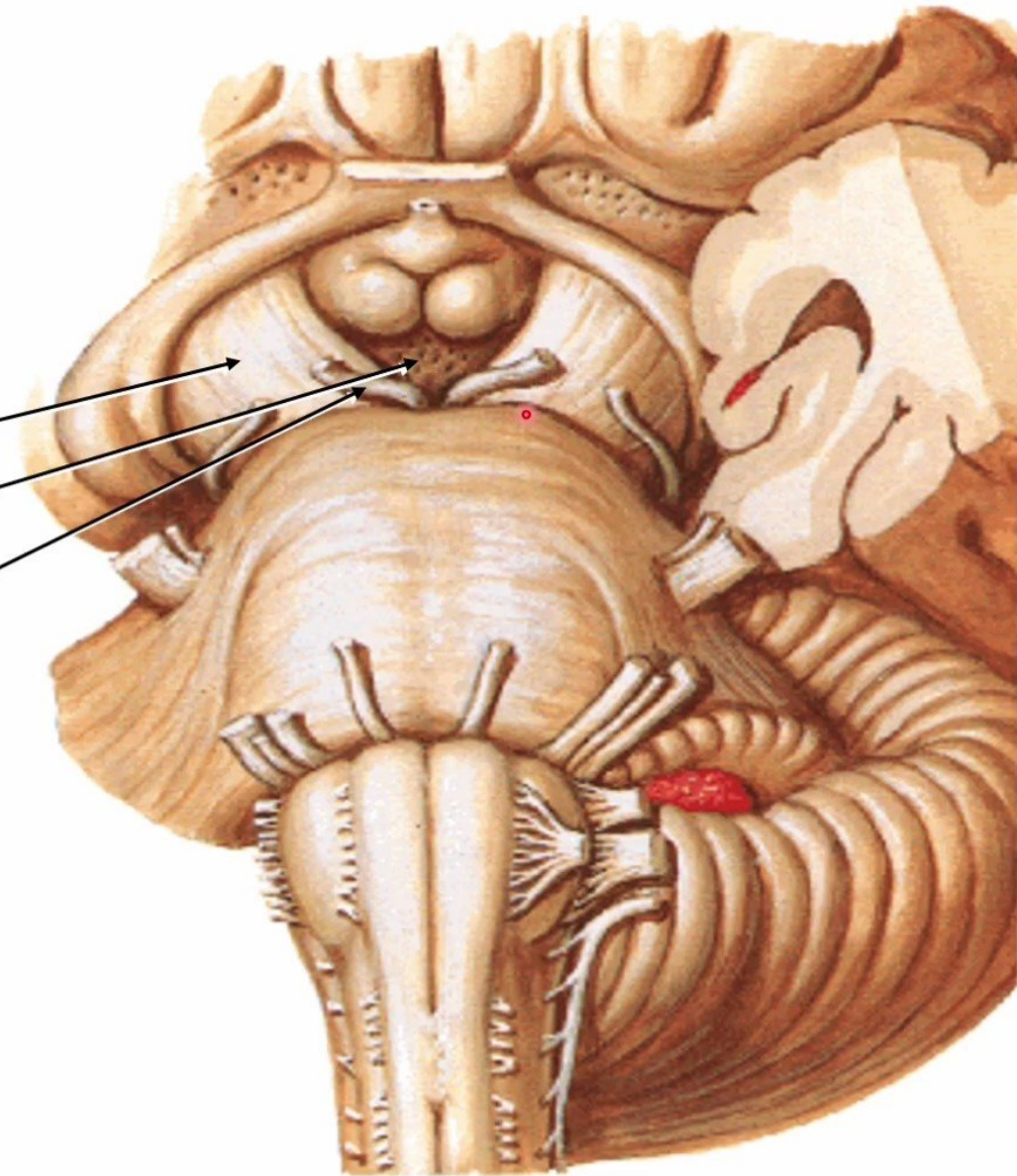
The anterior view of midbrain

The interpeduncular fossa

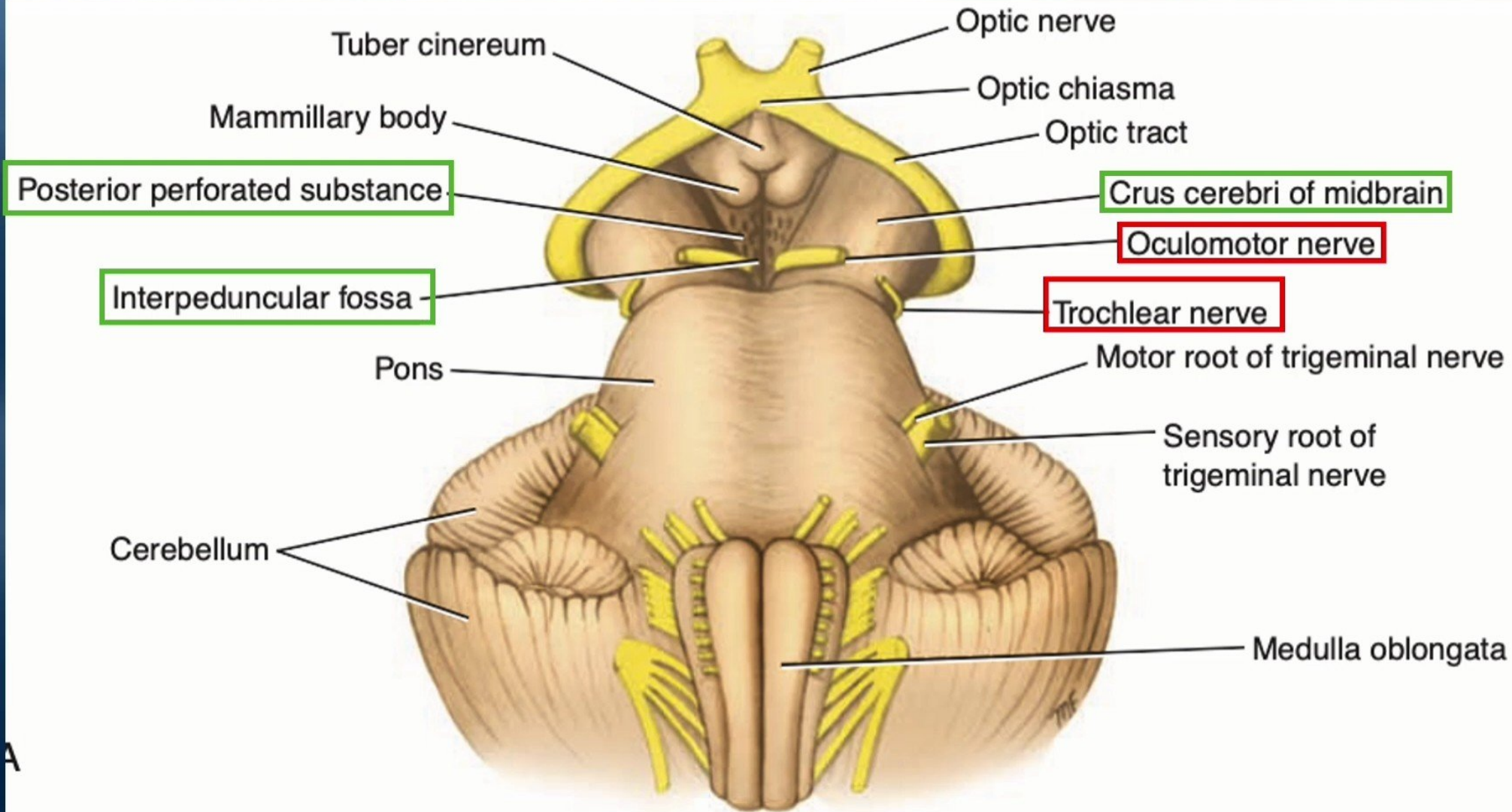
- It is found on the **anterior** aspect of the **midbrain** a deep depression in the midline.
- It is bounded on either side by the **crus cerebri**.
- Many **small blood vessels perforate** the floor of the interpeduncular fossa. therefore this region is termed the **posterior perforated substance**.

The oculomotor nerve

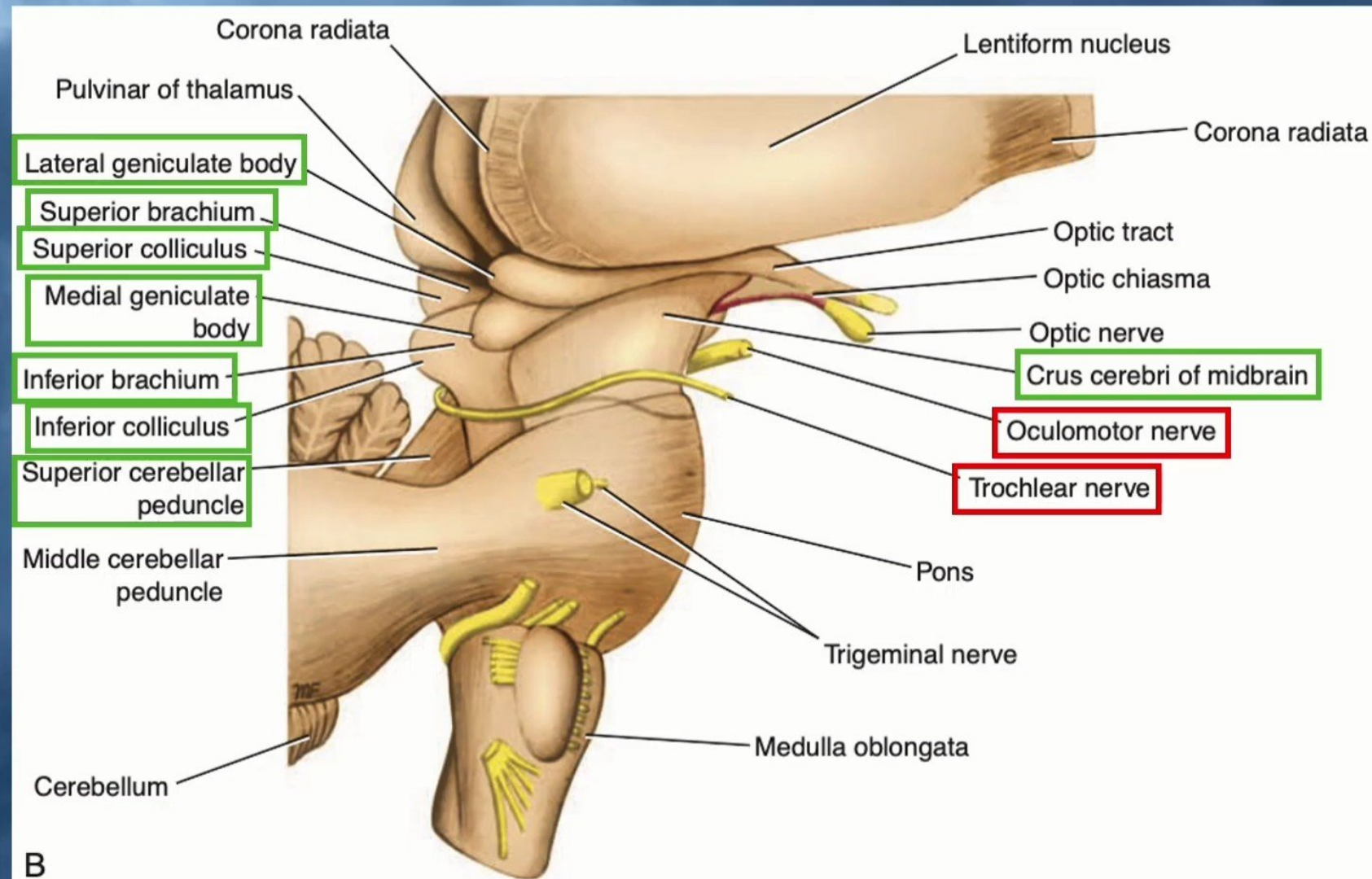
- It emerges from a groove on the medial side of the crus cerebri.
- It passes forward in the lateral wall of the **cavernous sinus**.



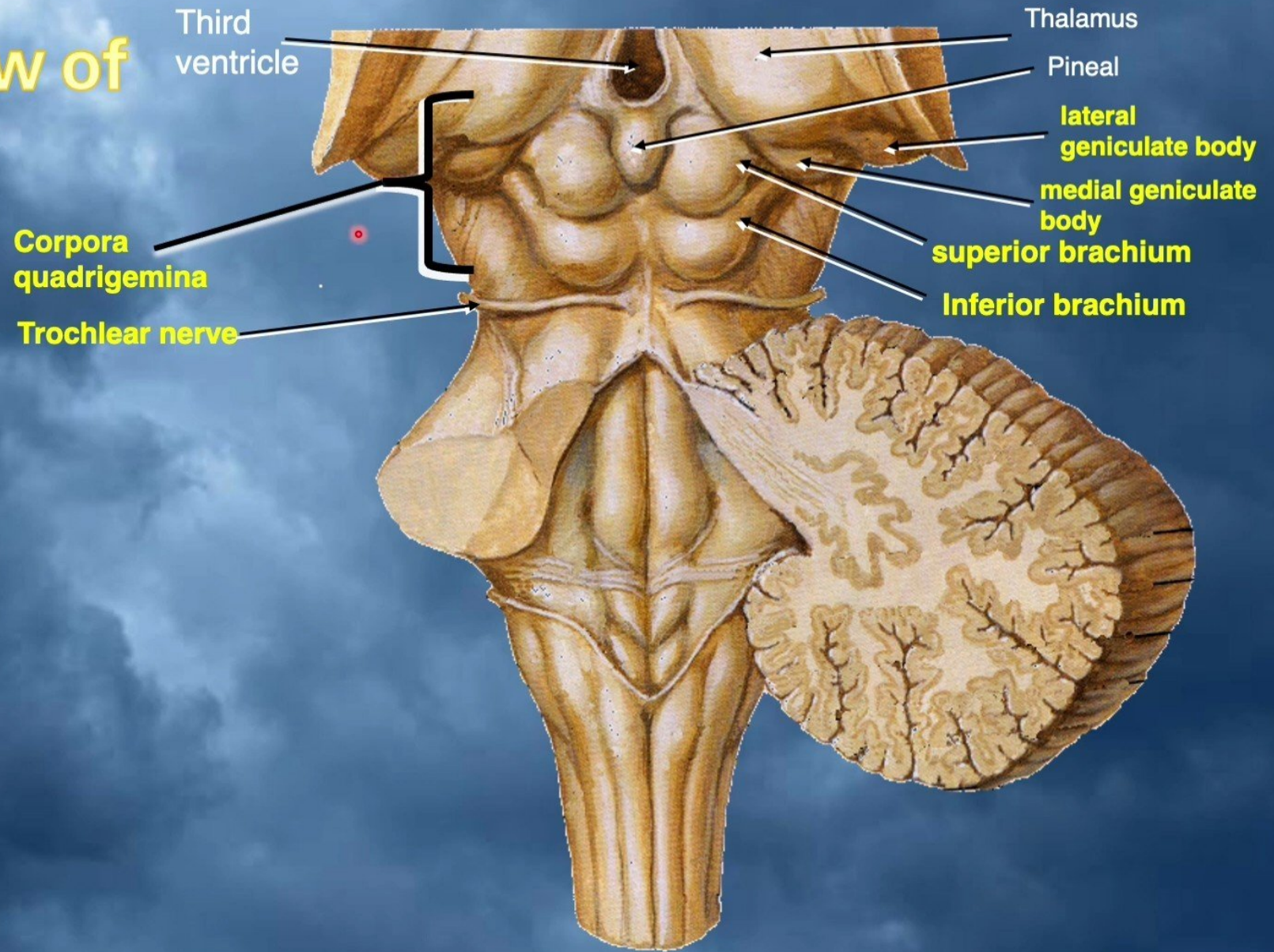
Anterior view of the midbrain



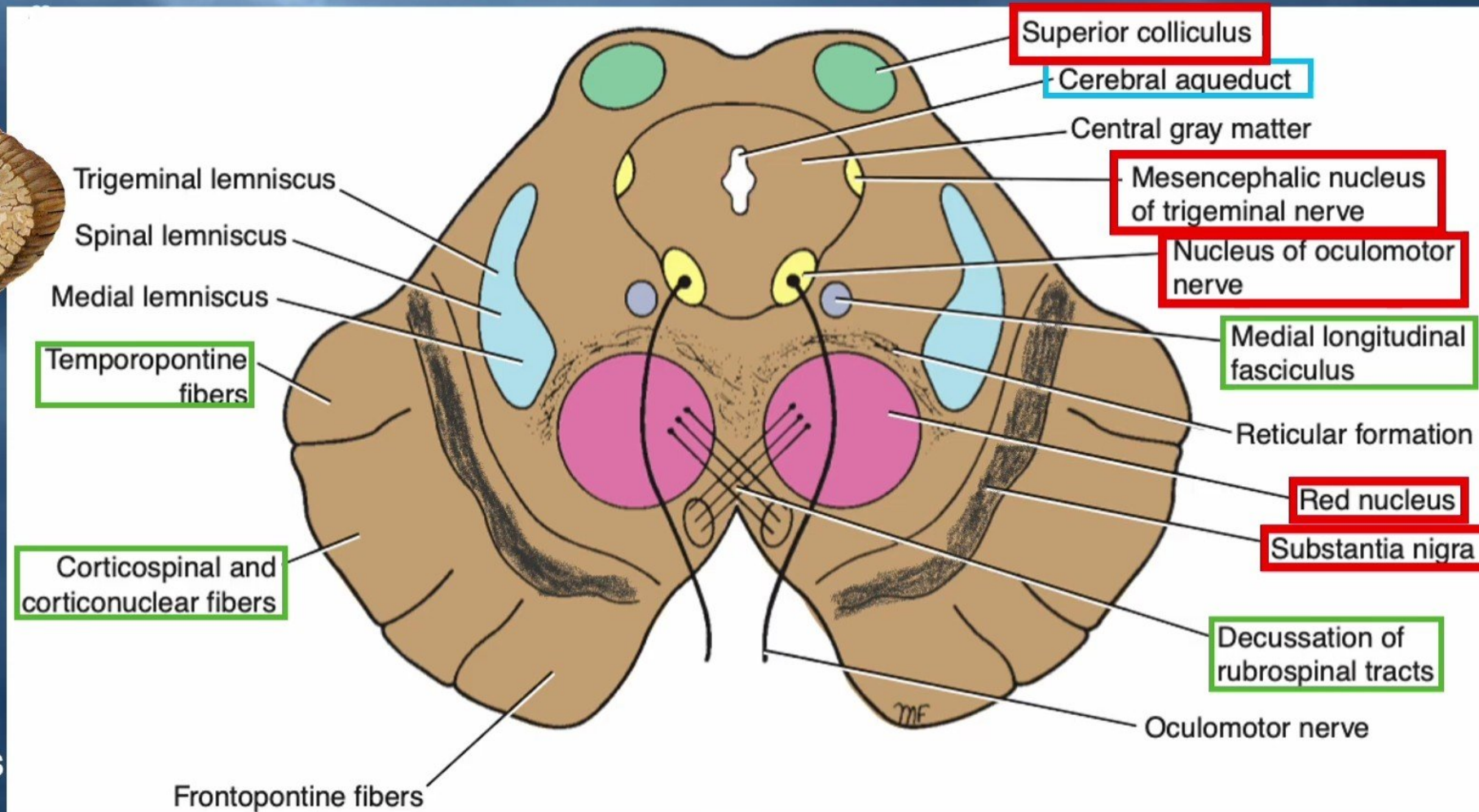
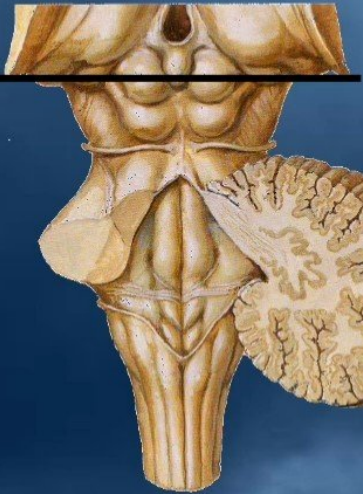
Lateral view of the midbrain



Posterior view of the midbrain



Transverse section at the level of the superior colliculus



- ✓ Level.
- ✓ Cavity.
- ✓ Nuclui.
- ✓ Motor tracts.
- ✓ Sensory tracts

The red nucleus

- It is a rounded mass of gray matter situated between the cerebral aqueduct and the substantia nigra. Why is it called red nucleus?

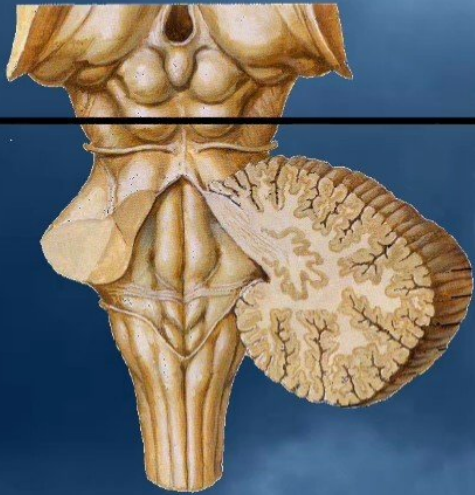
Afferent fibers reach the red nucleus from:

- (1) The cerebral cortex through the corticospinal fibers.
- (2) The cerebellum through the superior cerebellar peduncle, and
- (3) The lentiform nucleus, subthalamic and hypothalamic nuclei, substantia nigra, and spinal cord.

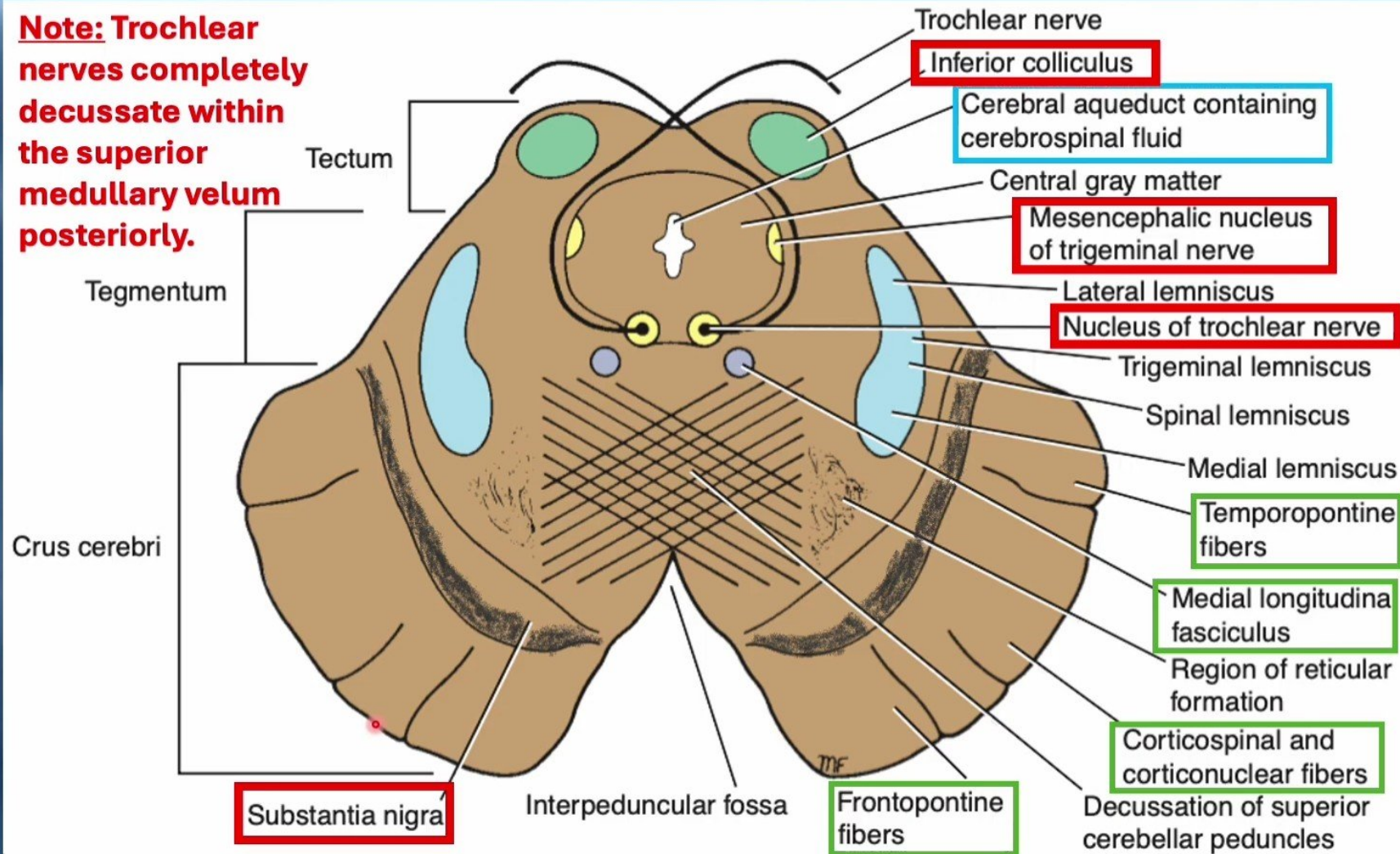
Efferent fibers leave the red nucleus and pass to:

- (1) The spinal cord through the rubrospinal tract (as this tract descends, it decussates).
- (2) The reticular formation through the rubroreticular tract.
- (3) The thalamus.
- (4) The substantia nigra.

Transverse sections of the midbrain at the level of the inferior colliculus



Note: Trochlear nerves completely decussate within the superior medullary velum posteriorly.

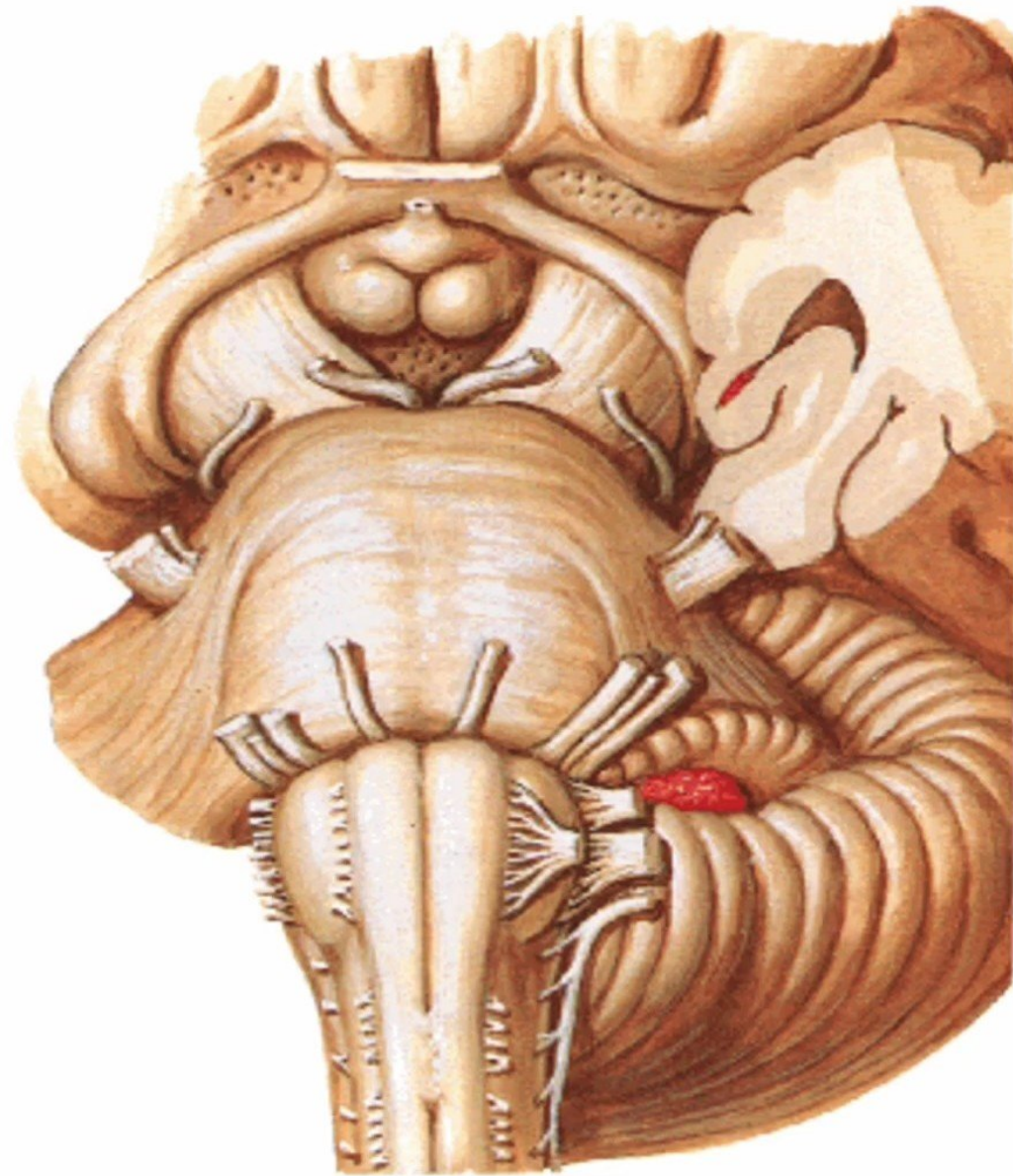


- ✓ Level.
- ✓ Cavity.
- ✓ Nuclui.
- ✓ Motor tracts.
- ✓ Sensory tracts.

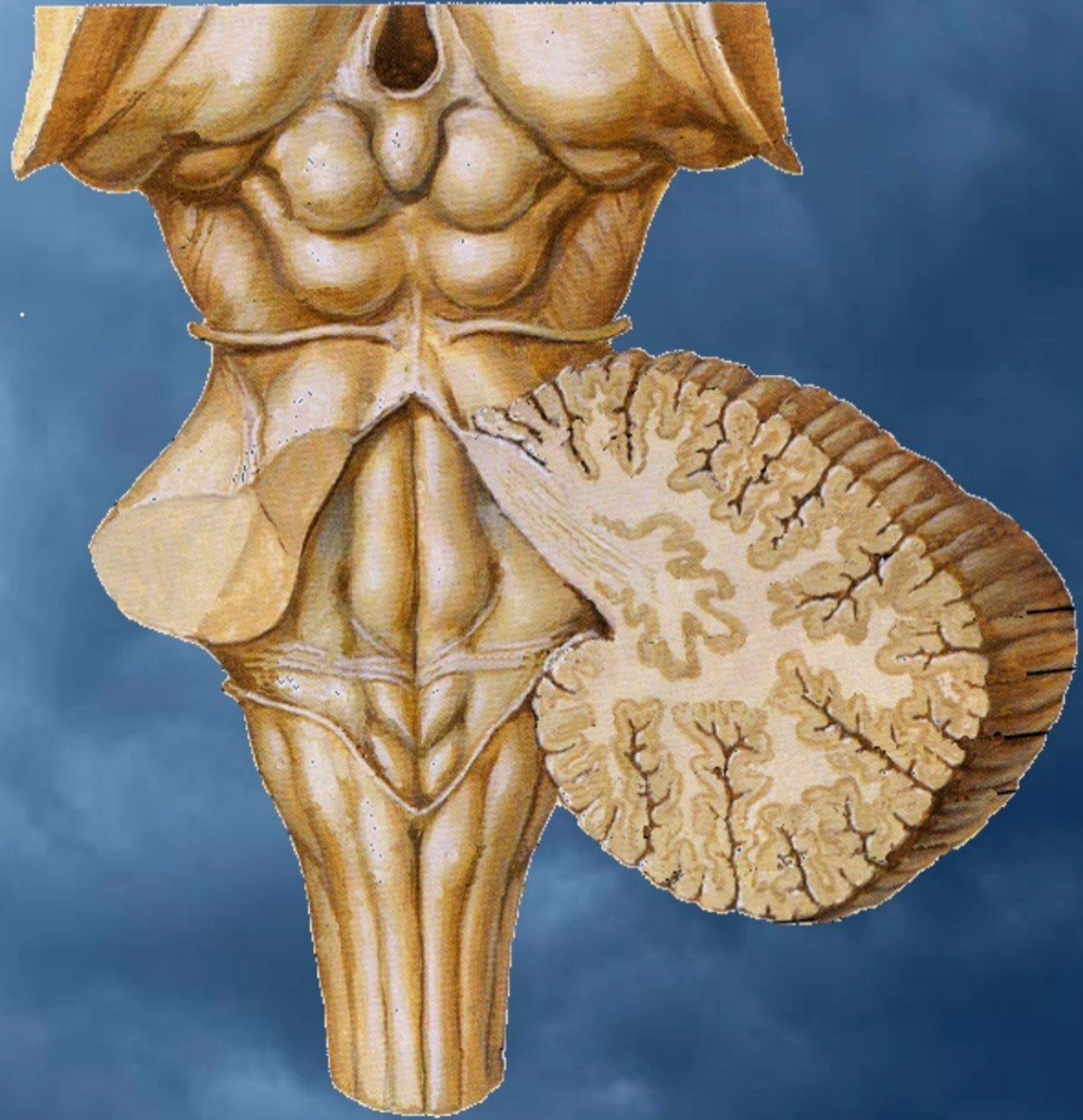
Comparison of Two Levels of the Midbrain Showing the Major Structures at Each Level^a

Level	Cavity	Nuclei	Motor Tract	Sensory Tracts
Inferior colliculi	Cerebral aqueduct	Inferior colliculus, substantia nigra, trochlear nucleus, mesencephalic nuclei of cranial nerve V	Corticospinal and corticonuclear tracts, temporo-pontine, fronto-pontine, medial longitudinal fasciculus	Lateral, trigeminal, spinal, and medial lemnisci; decussation of superior cerebellar peduncles
Superior colliculi	Cerebral aqueduct	Superior colliculus, substantia nigra, oculomotor nucleus, Edinger-Westphal nucleus, red nucleus, mesencephalic nucleus of cranial nerve V	Corticospinal and corticonuclear tracts, temporo-pontine, fronto-pontine, medial longitudinal fasciculus, decussation of rubrospinal tract	Trigeminal, spinal, and medial lemnisci

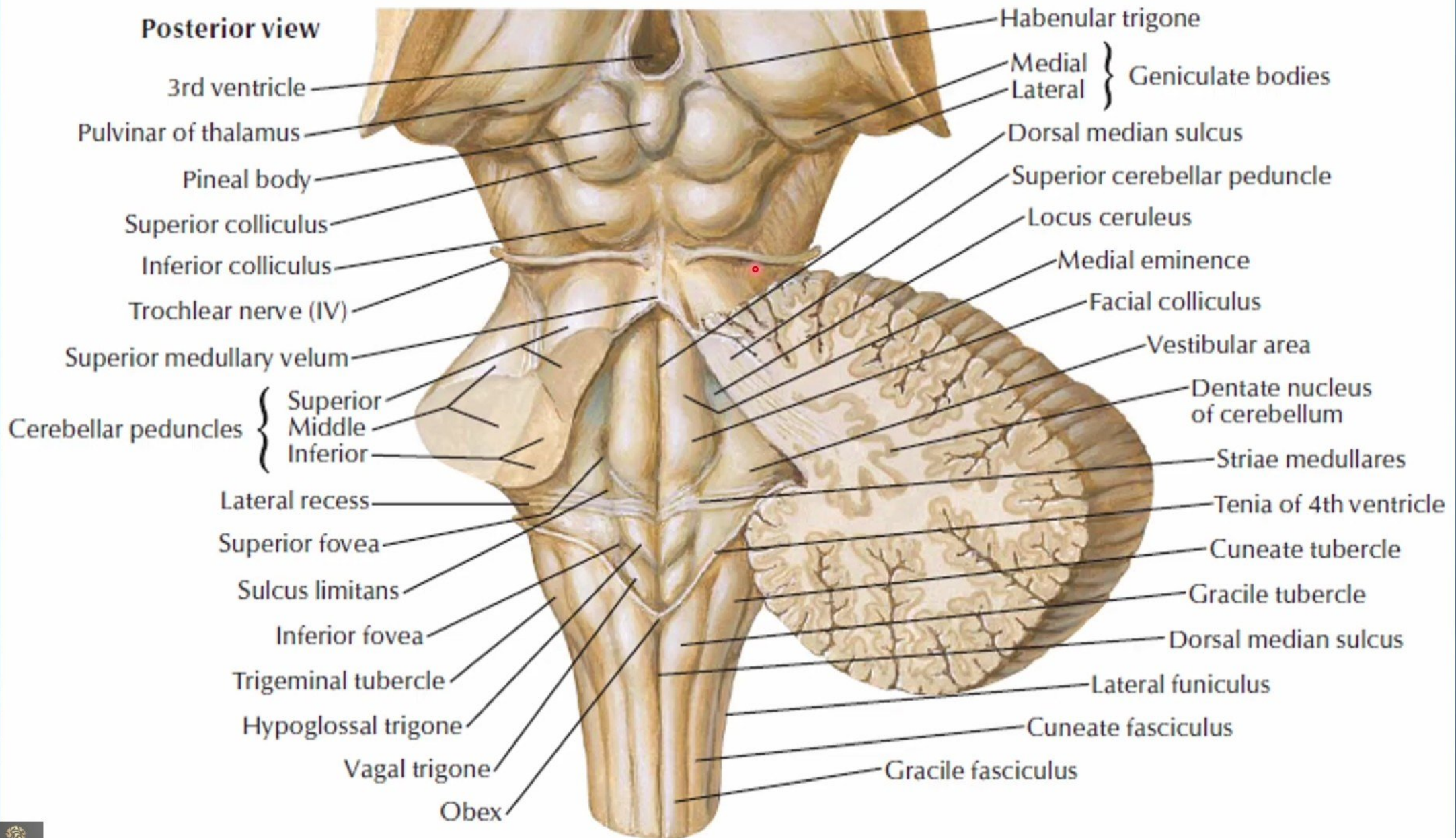
Summary of Anterior view of the Brain Stem



Summary of Posterior view of the Brain Stem

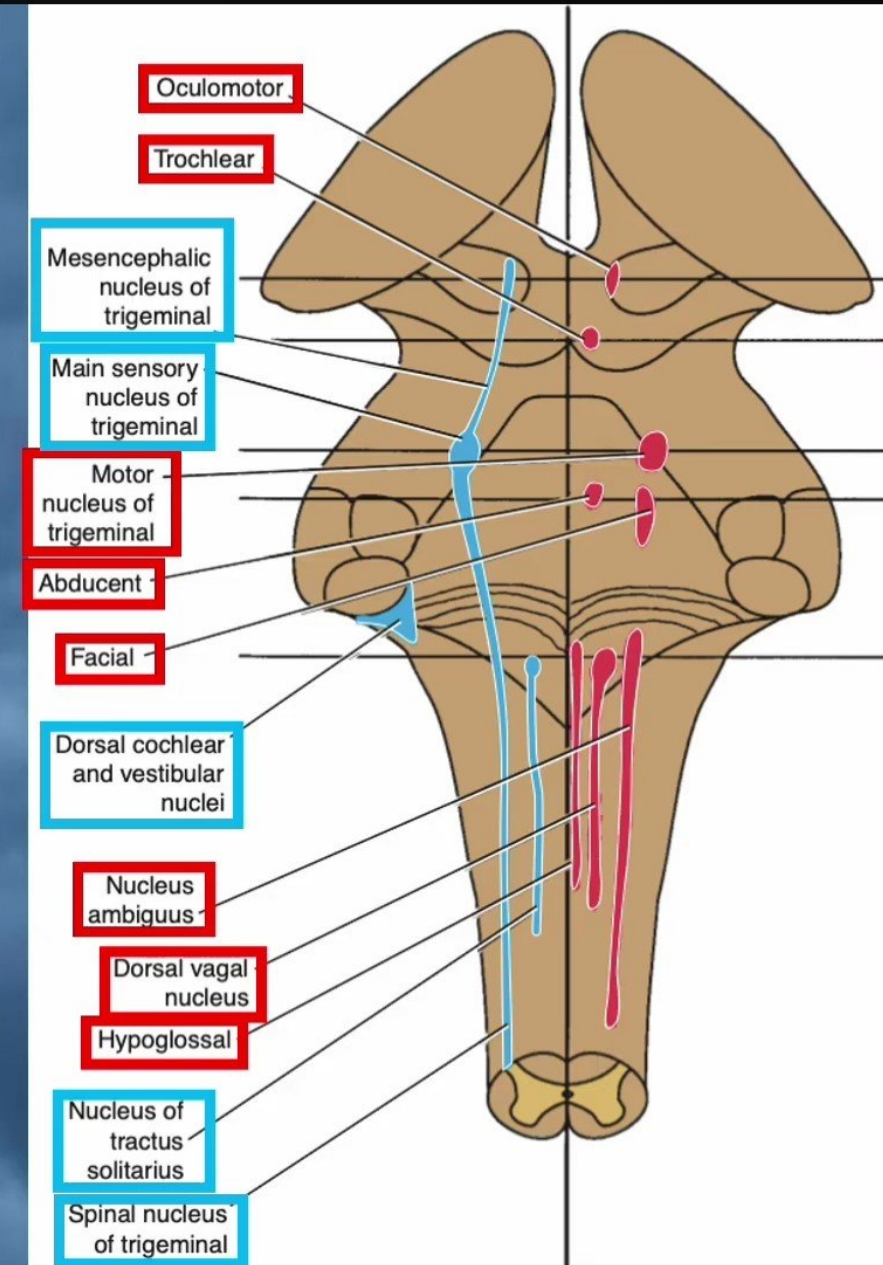


Posterior view



Summary of Cranial nerve nuclei in brain stem

- Cross sections of Brain stem.
- The motor nuclei are in red and
- The sensory nuclei in blue.



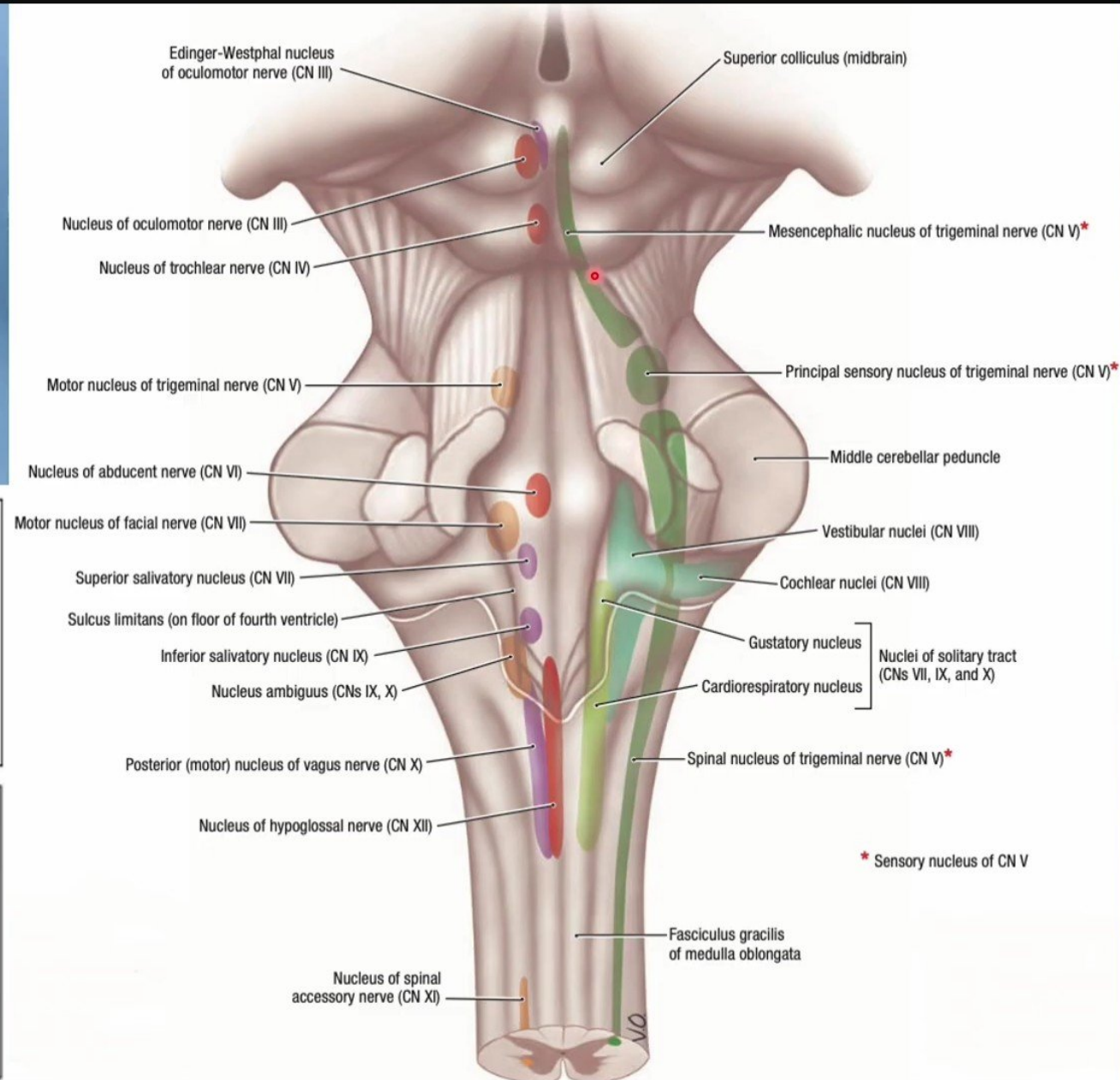
Cranial nerve nuclei

Sensory Nuclei:

- Somatic (general) sensory
- Special sensory
- Visceral sensory

Motor Nuclei:

- Somatic motor (including
- Branchial motor)
- Visceral motor (Parasympathetic)



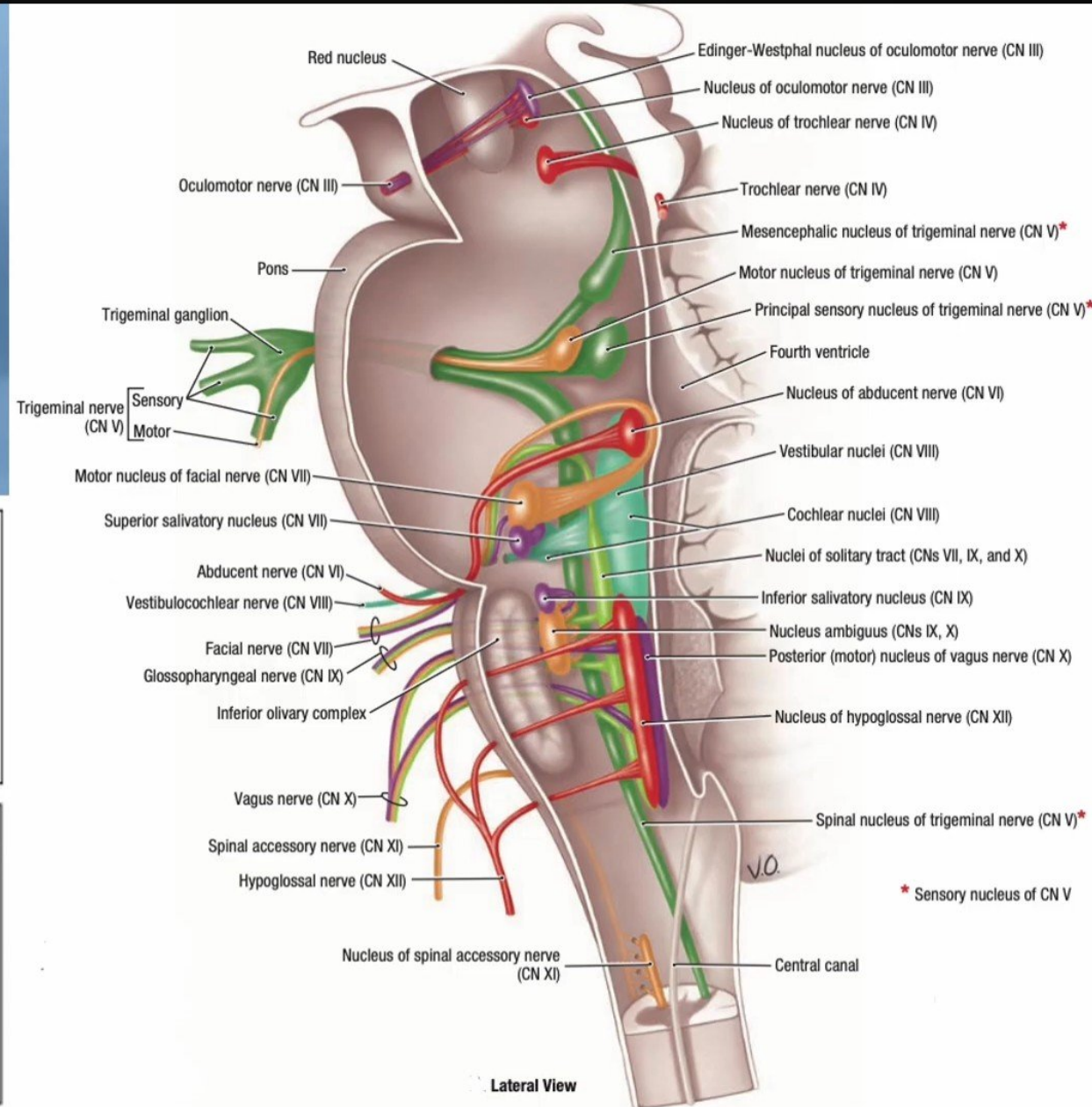
Summary of Cranial nerve nuclei in brain stem

Sensory Nuclei:

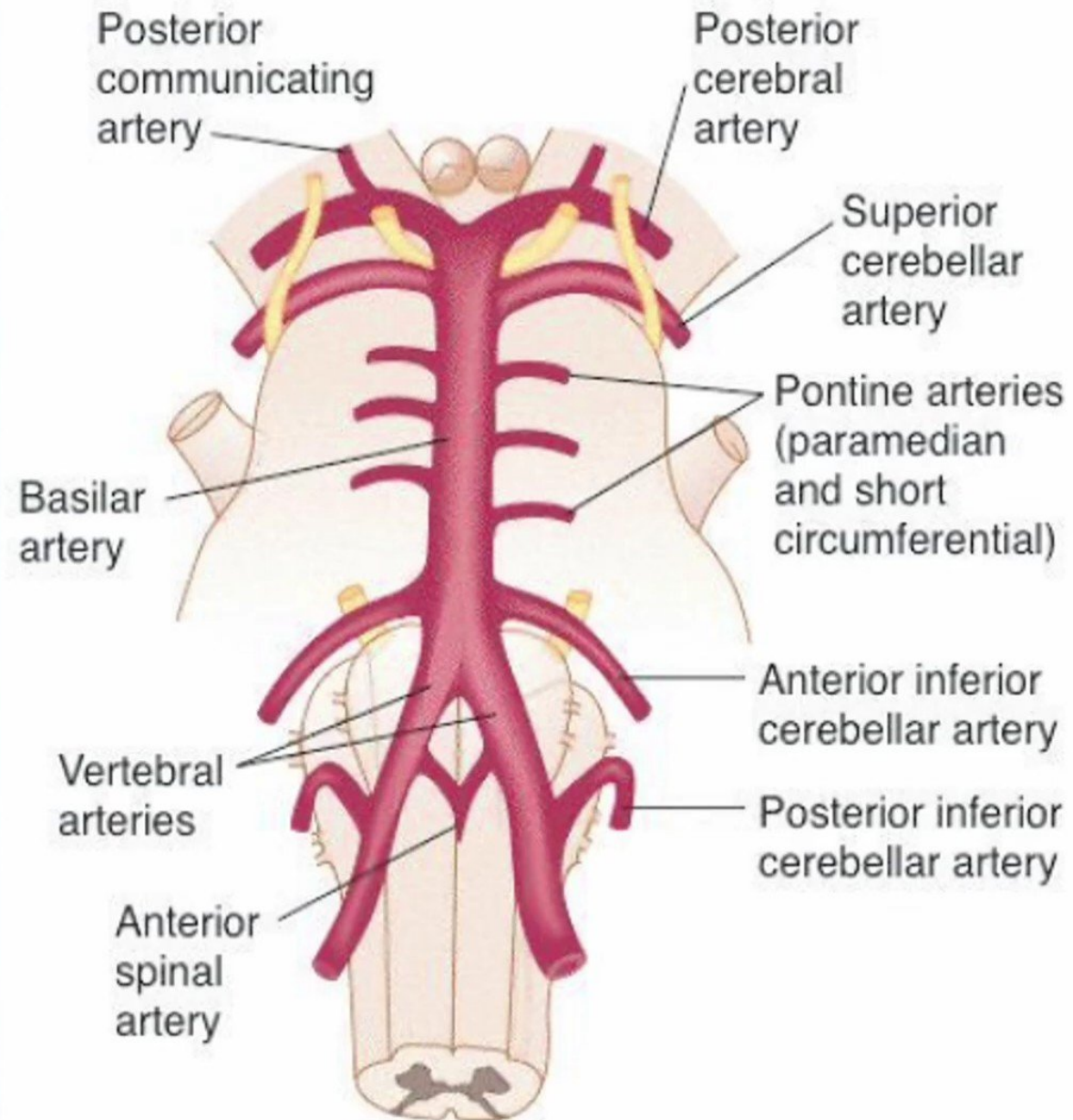
- Somatic (general) sensory
- Special sensory
- Visceral sensory

Motor Nuclei:

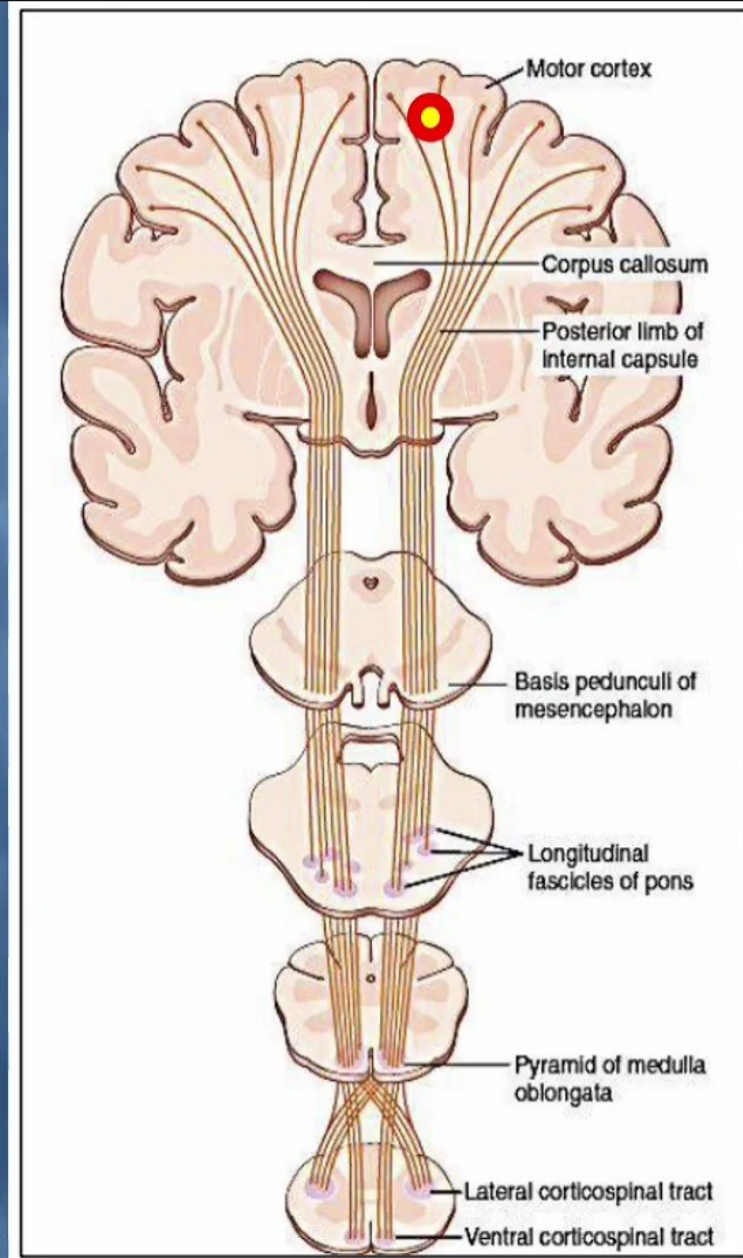
- Somatic motor (including
- Branchial motor)
- Visceral motor (Parasympathetic)



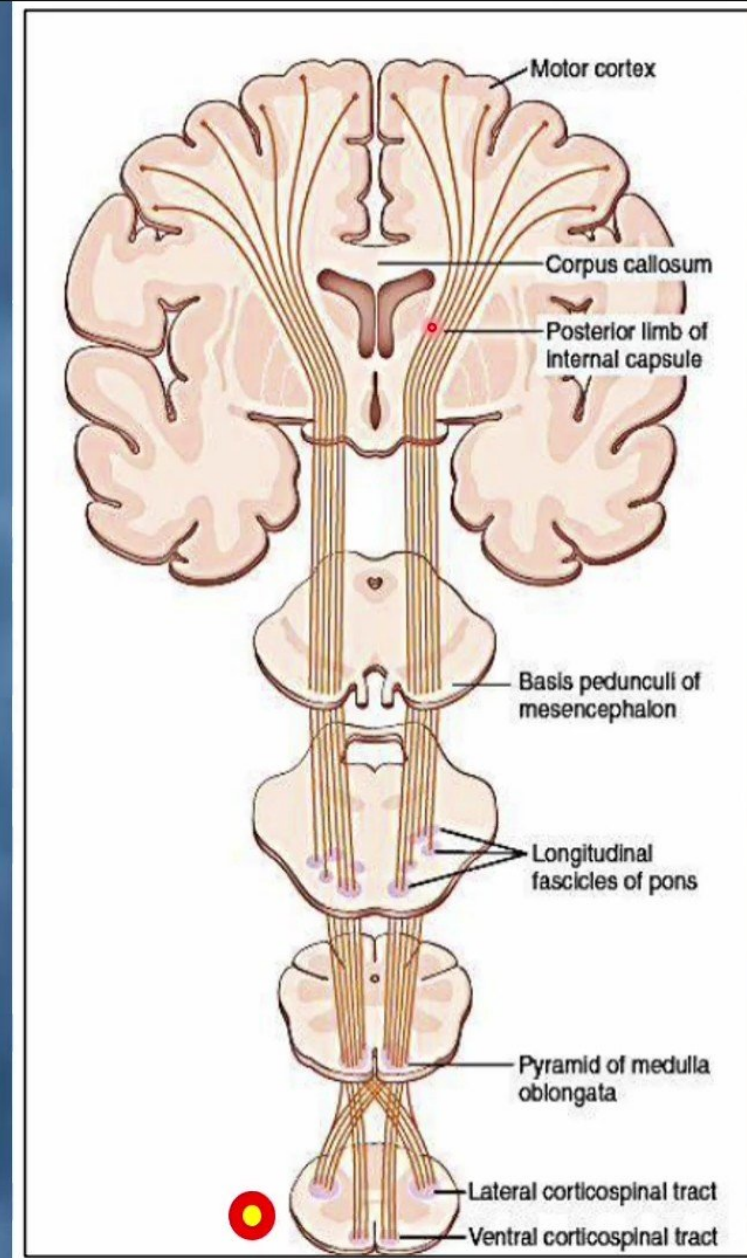
Principal arteries of the brain stem (Ventral View)



Corticospinal (pyramidal) tract



Corticospinal (pyramidal) tract



Thanks